

Lorenzo Gentile

🏠 Copenhagen, Denmark
🌐 <https://lorenzogentile404.github.io/>
🌐 <https://www.linkedin.com/in/lorenzogentile404/>
🌐 <https://www.youtube.com/@lorenzogentile404>
🌐 <https://orcid.org/0009-0009-5715-6368>



Work Experience

- 09/2023 - **Research Engineer at Consensys - Remote**
Researching and developing the arithmetization for a zkEVM (Zero-Knowledge Ethereum Virtual Machine) and a RISC-V zkVM (Zero-Knowledge Virtual Machine) as part of Linea.
- 06/2023 - 07/2023 **Lecturer at IT University of Copenhagen - Computer Science Department, Copenhagen - Denmark**
Planning and conducting lectures, exercise classes, assignments and examination activities for:
 - Applied Information Security (Summer 2023)
- 11/2019 - 01/2023 **PhD Student at IT University of Copenhagen - Computer Science Department, Copenhagen - Denmark**
Thesis: *Expanding Blockchain Horizons through Privacy-Preserving Computation* (https://lorenzogentile404.github.io/files/phd_thesis.pdf)
Advisor: Bernardo David

Research focused on cryptographic protocols for multiparty computation and blockchain applications.
- 03/2022 - 08/2022 **Scientific Partner at Concordium, Copenhagen - Denmark**
Collaboration with Concordium on multiparty computation as part of IT University of Copenhagen cryptography's research group.
- 03/2022 - 08/2022 **Guest PhD Student at Technische Universität Darmstadt - Computer Science Department, Darmstadt - Germany**
Advisor: Sebastian Faust

Pursue my PhD research project in a new environment supported by a scholarship from Collaborative Research Center 1119 CROSSING.
- 10/2018 - 10/2019 **Research Assistant at IT University of Copenhagen - Business IT Department, Copenhagen - Denmark**
Contribution to the analysis and the design of blockchain based systems, implementation of solutions to conduct research activities and to the organization of educational activities.
- 01/2018 - 10/2019 **Freelance Software Engineer**
Contribution to projects with a focus on statistical and simulation software.
- 04/2018 - 09/2018 **Technical Consultant at Observatory on Blockchain and Distributed Ledger - School of Management of Politecnico di Milano, Milan - Italy**
Contribution to a comparative analysis of the state-of-the-art blockchain projects and to the creation of educational contents.

- 09/2015 – 09/2017

■ **Research Collaborator at Politecnico di Milano – Dipartimento di Matematica, Milan – Italy**
 Investigation related to the usage of blockchain in the insurance field.
- 06/2008

■ **Analyst and Developer at MOXOFF S.p.A. (spinoff Politecnico di Milano), Milan – Italy**
 Research, design and development of mathematical, numerical and statistical methods for industrial application.

■ **Software Developer at Politecnico di Milano – Dipartimento di Elettrotecnica, Milan – Italy**
 Development of a home automation software using Zigbee devices.

Education

- 11/2019 – 01/2023

■ **Doctor of Philosophy – PhD, Computer Science at IT University of Copenhagen, Copenhagen – Denmark**
 Thesis: *Expanding Blockchain Horizons through Privacy-Preserving Computation* (https://lorenzogentile404.github.io/files/phd_thesis.pdf)
 Advisor: Bernardo David

Research focused on cryptographic protocols for multiparty computation and blockchain applications.
- 09/2017 – 10/2017

■ **Entrepreneurship at Draper University, San Mateo – California**
 Program that provided practical startup skills, access to industry professionals, a network of entrepreneurs, and mentorship from investors.
- 10/2012 – 07/2015

■ **M.Sc. in Engineering of Computing Systems at Politecnico di Milano, Milan – Italy**
 Thesis: *A comparative study of mechanisms for Sponsored Search Auctions* (https://www.politesi.polimi.it/bitstream/10589/108785/3/2015_07_Gentile_Gentile.pdf)
 Advisor: Nicola Gatti

Studies focused on Mathematical Optimization, Mechanism Design and Machine Learning.
- 09/2009 – 09/2012

■ **B.Sc. in Engineering of Computing Systems at Politecnico di Milano, Milan – Italy**
 Final project: *a Java implementation of the Carcassonne board game*
- 09/2004 – 07/2009

■ **Secondary School Diploma in Information Technology at Istituto di Istruzione Superiore Luigi Galvani, Milan – Italy**
 Thesis: *Zigbee Technology and its Applications*

Skills

- | | |
|------------------------|---|
| Languages | ■ English (full professional proficiency), Italian (native or bilingual proficiency), Danish (elementary proficiency). |
| Knowledge areas | ■ Cryptography, Software Engineering, Mathematical Optimization, Game Theory, Computer Security, Machine Learning. |
| Tools and Technologies | ■ Java, Python, Solidity, Rust, Go, Zig, R, GNU Octave, Haskell, C, C++, SQL, L ^A T _E X, Ethereum. |

Selected Publications

- 1 Bégassat, O., Bojarski-Nasretdinov, F., **Gentile, L.**, Pearce, D., Chatelain, A., & Belling, A. (2026). Linea ZK-EVM specification [<https://zenodo.org/records/20200906> (published through Consensus)].
- 2 Brorsson, J., David, B., **Gentile, L.**, Pagnin, E., & Wagner, P. S. (2023). PAPR: Publicly Auditable Privacy Revocation for Anonymous Credentials [<https://eprint.iacr.org/2023/137> (published to CT-RSA 2023)].
- 3 Baum, C., Chiang, J. H.-y., David, B., Frederiksen, T. K., & **Gentile, L.** (2021). SoK: Mitigation of Front-running in Decentralized Finance [<https://ia.cr/2021/1628> (published to DeFi 2022 - FC 2022 workshop)].
- 4 David, B., **Gentile, L.**, & Pourpouneh, M. (2021). FAST: Fair Auctions via Secret Transactions [<https://eprint.iacr.org/2021/264> (published to ACNS 2022)].

Selected Teaching and Dissemination Activities

- | | |
|-------------------------|--|
| 06/2023 - 07/2023 | ■ Applied Information Security - IT University of Copenhagen, Copenhagen - Denmark
This is a hands-on course that teaches the basic principles of computer security, where students have the chance to gain in-depth experience with cyberattacks, and how to prevent them. |
| 06/2022 | ■ 20th International Conference on Applied Cryptography and Network Security (ACNS 2022), Rome - Italy
At ACNS, an annual conference focusing on current developments that advance the areas of applied cryptography and its application to systems and network security, I had the chance to present [4]. |
| 01/2021 - 06/2021 | ■ Cryptographic Computation and Blockchain - IT University of Copenhagen, Copenhagen - Denmark
This course introduces basic concepts and techniques for designing and analysing cryptographic protocols with a focus on privacy preserving computation and blockchain protocols. It covers both the main constructions of such protocols and the theoretical models used for proving their security. |
| 08/202(01) - 01/202(12) | ■ Security 1 - IT University of Copenhagen, Copenhagen - Denmark
This is an introductory course on information security. The course focuses on introductory aspects of analysis, design and implementation of secure software. |